

2.6 Shells and Virtual Consoles

2.6.1 Shells

A shell is a command interpreter that reads the command you type and, according to the command, performs the requested operation. A command that you type must be recognised by the command interpreter. The most commonly used shell in Linux is the “bash” shell. It is available in all versions of Linux, and is an offshoot from Unix. Like most things in Linux, “bash,” too, is a contraction, and is derived from “Bourne again shell.” The bash shell can distinguish between alphabet case types. Hence “*LS*” is not the same as “*ls*”.

Commands that are not part of a shell are called “external commands,” and relate to programs. Shell scripts can also be invoked by typing a command, which can be external—a shell script is nothing but a set of commands stored in a “script,” or a file consisting of several commands that can be executed in a single go.

As a new user, you may not see an immediate need for shell programming. But at the level of system administrators or database administrators, shell programming is very useful. There are several complex commands that will be difficult to type in every time they are needed. A script can also be run at regular intervals to monitor system performance or network traffic. Another area where shell scripts can be of use is in backing up data. If data is stored in different locations, manually copying it to a backup location can be a tedious task—which can be simplified using a script. No wonder you see such a demand for shell programmers!

Apart from bash, other command interpreters include the C shell (csh), Korn shell (ksh), Bourne shell (sh), and more.

A command interpreter is a program that reads (rather, interprets) text commands typed by the user (or also from a file if it is a script file) and executes them.